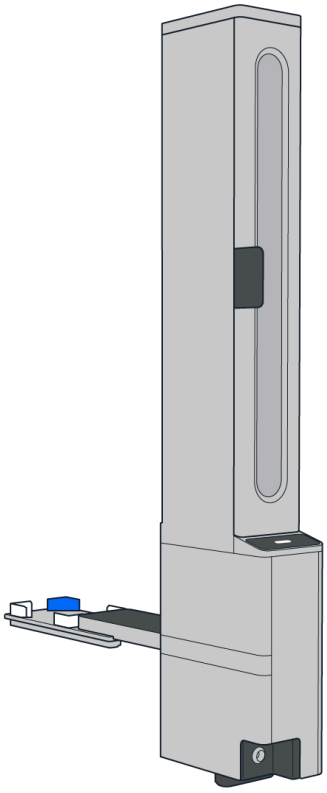
Stacker Retrofit Kit

Instruction Manual



Opentrons Labworks Inc.
November 2025

Table of Contents

Product Description
Affected Serial Numbers
Upgrade Kit Elements
    Kit Part Number
    Box Contents
Kit Contents Overview
    Rear Panel Board
    Window-to-Flex Retrofit Board
    Window Contact Block
Before You Begin
    Electrical
    Tools
    Workspace
Retrofit Instructions
    Part 1: Preparing the Robot and Stacker
    Part 2: Replacing the Rear Panel Board
    Part 3: Attaching the Window Panel Block
    Part 4: Retrofitting the Stacker Window Panel
    Final Steps
Retrofit Procedure Resources

Product Description

This retrofit kit adds electronic hardware to Flex robots that prevents the HEPA/UV module from operating if the Stacker is improperly installed. It is required for most models manufactured before September 2025, or as indicated by the robot's serial number. Newer robots include these upgrades and do not need retrofitting. Only qualified Opentrons personnel should install the retrofit kit.

Affected Serial Numbers

Any Flex with a serial number version that includes **A10** or **A20** *requires* a Stacker retrofit kit, even if you don't have the HEPA/UV module installed. Robots with serial number version **A30** (or higher) are fully compatible with the Stacker; the retrofit kit is not required. The kit adds new firmware and hardware to your Flex which:

- Allows early model robots to recognize and communicate with the Stacker.
- Prevents the HEPA/UV module from operating if the Stacker is improperly installed.

The serial number can be found on the certification sticker on the back of each Flex near the on/off switch, or under the touchscreen, or in the Opentrons App under the **Advanced** tab.

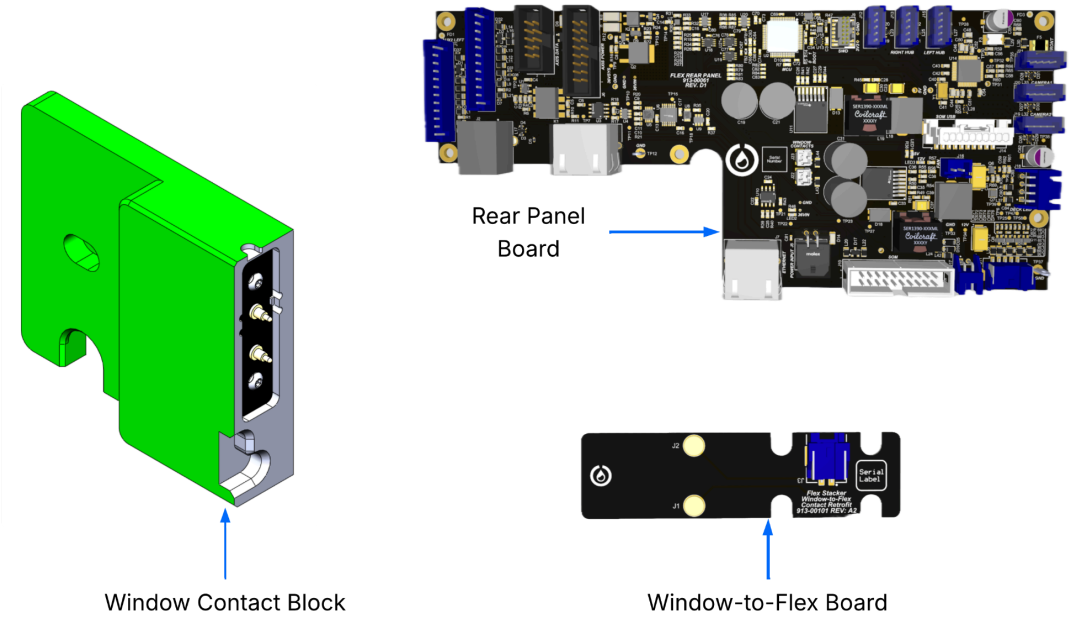


> **Note:**
> The retrofit kit must be installed by an Opentrons field service technician only.

Compare these serial number versions to the serial number on your robot to determine if it needs the Stacker retrofit kit installed.

Version	Example	Retrofit required
A10	FLX**A10**20240201234	Yes
A20	FLX**A20**20250321003	Yes
A30	FLX**A30**20250908004	No

Upgrade Kit Elements



Kit Part Number

The part number for the Stacker retrofit kit is 999-00270.

Box Contents

(1) Rear panel board
(1) Window contact block
(1) Window-to-Flex contact board

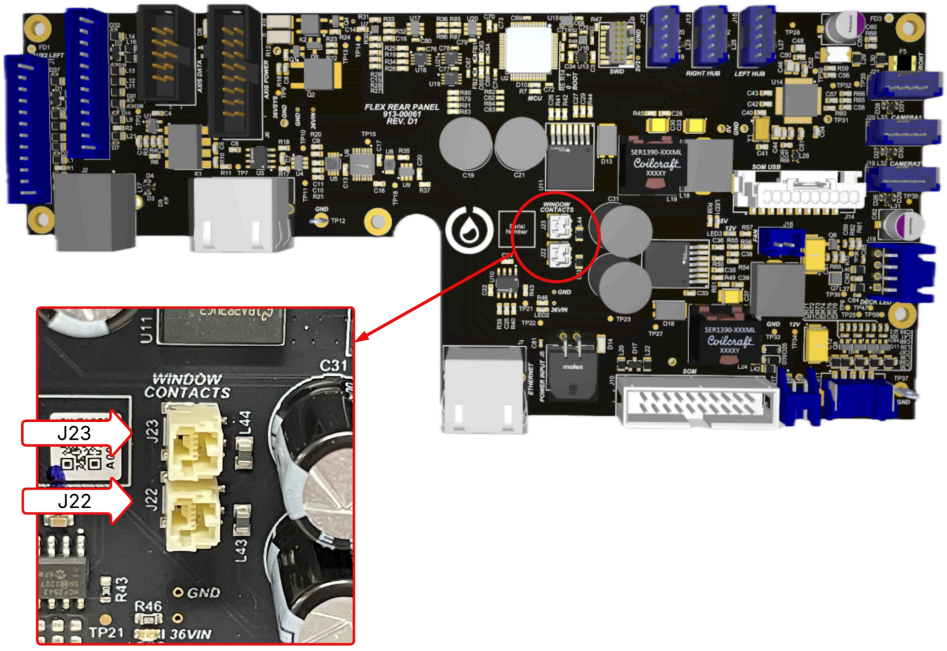
(1) T8 Torx screwdriver
(1) T10 Torx screwdriver
Replacement screws

Kit Contents Overview

The retrofit kit includes new printed circuit board assemblies (PCBAs) that look similar to the boards already installed in a Flex. This section highlights what's different on the new boards and how to distinguish them from older, previously installed PCBAs.

Rear Panel Board

The rear panel board in the retrofit kit includes two new connection points, J22 and J23 (shown below).



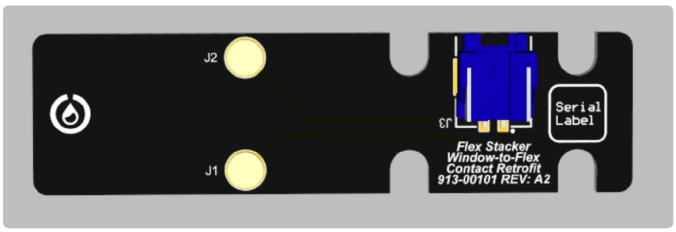
In the kit, this board may ship with one of its 2-pin connectors (J22 or J23) shorted by a jumper. During the retrofit, leave the jumper in place and connect the wires from the window contact block to whatever connector is available.

The rear panel board uses the same part number (913-00061) as its predecessors, but the revision ID changes to "REV. D1". These identifying marks are printed on each new board as shown below.

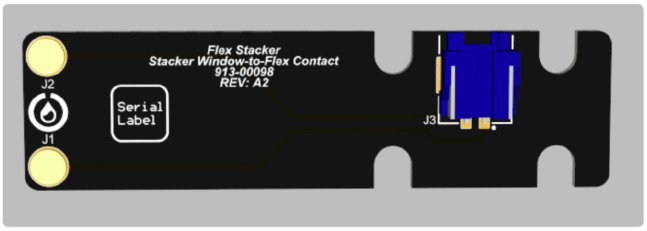
Flex Rear Panel
913-00061
REV. D1

Window-to-Flex Retrofit Board

The retrofit kit also includes the window-to-Flex retrofit board. It replaces the original board on the Stacker window panel, which you'll remove in Part 3 of this retrofit procedure.



Retrofit Window Panel PCBA



Original Window Panel PCBA

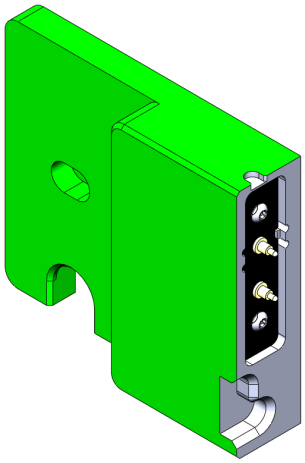
The window contact boards are almost identical in appearance and can be identified by these differences:

- **Part numbers**:
  - Original board: 913-00098
  - Replacement board: 913-00101

- **Contact point location**: On the replacement board, the contact points, labeled J1 and J2, are located near the middle of the board.

Window Contact Block

Contact points on the window panel enable the robot to detect if the lower side window is installed.



Before You Begin

Before performing any work on Flex, review the following safety information and recommendations.

Electrical

Turn off the power and remove the power cord from the robot and any attached modules before you begin. Sometimes you may need to turn on the power to check LED status lights on a circuit board or to test cable connections. In these cases, always turn the power off after verifying status lights or when removing/attaching boards and cables. Never remove, disconnect, and/or attach controller boards, cables, and cover panels while the power is on.

Tools

This retrofit procedure requires the following tools:

- 2.5 mm hex screwdriver
- 3 mm hex screwdriver
- T8 Torx screwdriver
- T10 Torx screwdriver

Workspace

Detach any instruments from the gantry and remove deck-mounted modules from the working area of the robot. Removing these items helps prevent accidental damage to Flex and gives you more room to work. You'll also need access to the back of the robot to reach the electronics bay cover and rear panel board.

Retrofit Instructions

These instructions will guide you through the retrofitting process. When complete, your customer's Flex robot and Stacker module will incorporate new and current safety features. These features prevent an attached HEPA/UV module from operating if the Stacker window panel or the Stacker are improperly attached.

Part 1: Preparing the Robot and Stacker

In this section, you will prepare the Flex and Stacker for retrofitting.

1. Turn off the power to the Flex.

2. Disconnect all cables from the back of the robot (e.g., power cable, E-stop AUX cable, HEPA/UV Module AUX cable).

3. Remove the Stacker and User Kit from their respective boxes. This product ships in two separate boxes; one contains the module, the other includes a special User Kit. Discard packing material as required.

4. Remove deck-mounted modules and labware. You may also want to detach any pipettes or the gripper, though this is optional. The goal here is to create working space for yourself in and around the robot.

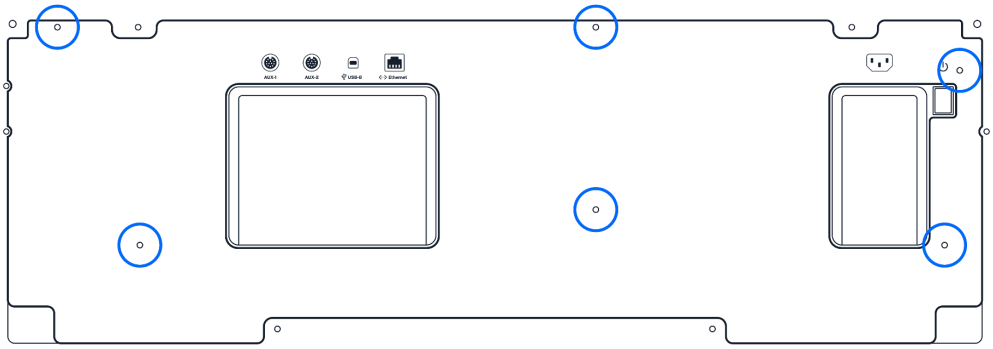
5. Using a 2.5 mm hex screwdriver, detach the lower right side window panel. Set it aside for safekeeping.

Part 2: Replacing the Rear Panel Board

In this section, you will replace the rear panel PCBA with the retrofit board from the kit. This board includes new connectors that complete the circuit between the window panel and the robot.

To replace the rear panel board:

6. Using a 3 mm hex screwdriver, remove the screws that hold the electronics bay cover in place and detach it.



The rear panel PCBA is behind the electronics bay cover and looks like this (without the retrofit):



7. Disconnect all the data and power cables from the rear panel board.

8. Using a 2.5 mm hex screwdriver, remove the screws holding the board to the robot and take out the board.

9. Hold the new board in place and reattach it with the screws you removed while removing it from the robot. Don't reattach the electronics bay cover or reconnect cables to the new board. You'll do that in the next section.

Part 3: Attaching the Window Panel Block

In this section, you'll attach the window panel block, connect it to the rear panel board, reconnect the cables, and put the electronics bay cover back on.

To attach the window panel block:

10. Using a 2.5 mm hex screwdriver, remove the right-side wire channel cover. This is a small plate located at the rear of the enclosure.



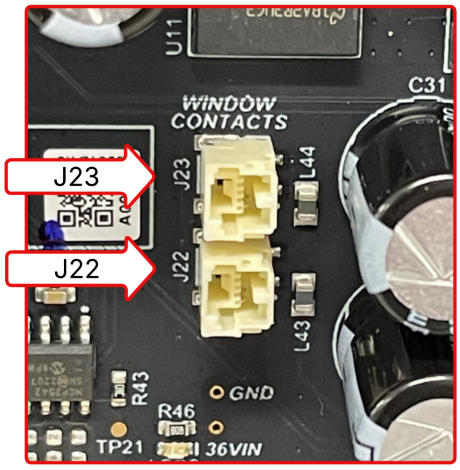


11. Take the window contact block from the retrofit kit. Route its connecting wire through the wire channel, then fasten the block to the robot using the same screw that held the wire channel cover in place.

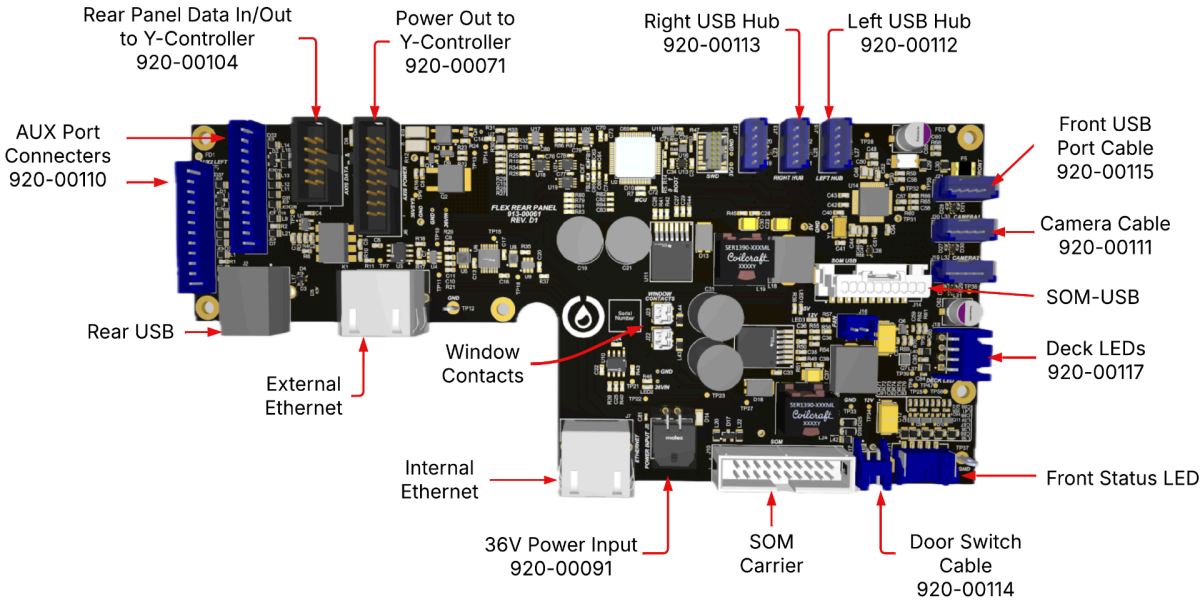
    The window contact block looks like this when attached.



12. Attach the loose end of the contact block wire to contact J22 or J23 on the rear panel board.



13. Reconnect the power and data cables to their original locations on the retrofit board. Use the illustration below as a guide. Match the part numbers on the cables to the connections as shown here.



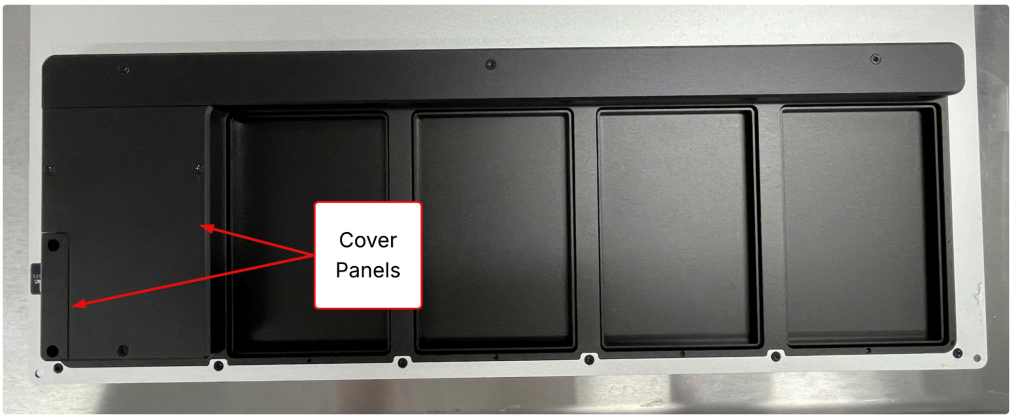
14. Reattach the electronics bay cover by holding it in place and securing it with the screws you removed earlier.

Part 4: Retrofitting the Stacker Window Panel

In this section, you will replace the window-to-Flex board in the Stacker window panel with a new PCBA from the kit.

To retrofit the Stacker window panel:

15. Place the panel on a flat surface with the interior side facing up, towards you. When attached to a Flex, this is the side that faces into the working area of the robot. Find the raised cover panels along one side of the panel.

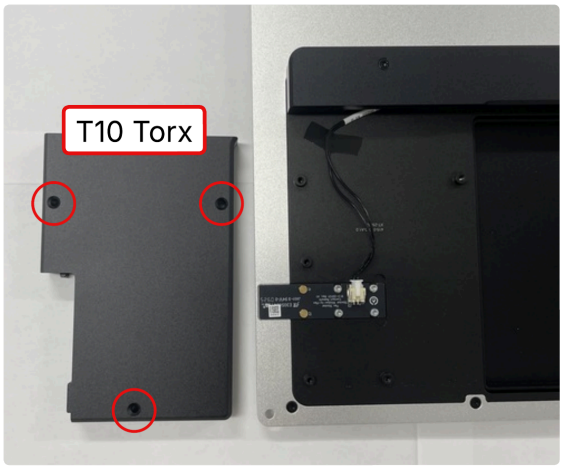


16. Using a 2.5 mm hex screwdriver, remove the 2 screws that hold the small cover plate subassembly in place and remove it. You will not reinstall this item.



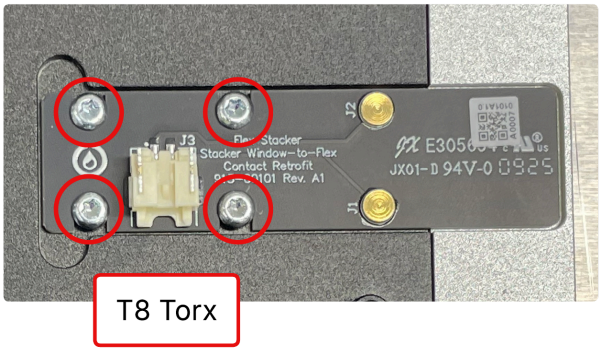
Note:
The small cover must be removed first, before the larger cover panel.

17. Using a T10 Torx screwdriver, remove the 3 screws that secure the larger cover plate to the window panel and detach the cover. Removing this panel fully exposes the circuit board.



18. Disconnect the cable that connects the small circuit board to the window panel.

19. Using a T8 Torx screwdriver, remove the 4 screws that fasten the small circuit board to the window panel. Replace it with the small board from the retrofit kit and secure it in place with the same screws you just removed.



20. Connect the cable from the window panel to the small circuit board.

21. Reattach the cover panel you removed in step 16 above. Do not replace the small cover panel. When complete, the board is partially exposed as shown below.



22. Attach the replacement side window panel to the robot. This completes the retrofit procedure.

Final Steps

The Stacker is ready to install after you complete the retrofit procedures. If you're not installing the Stacker right away, then reattach all the cables you removed from the Flex earlier. Refer to these resources for Stacker installation instructions:

- The printed Quickstart Guide that ships with the Stacker.
- The online Stacker Instruction Manual.
- The retrofit video linked below (starting at the 6:20 timestamp).

Retrofit Procedure Resources

This manual is a companion piece to these other retrofit guides.

Resource	Description
Retrofit video	The Flex Stacker Module Retrofit Video (11 minutes) shows you how to install and connect the upgraded boards in the kit. It also provides installation instructions for the Stacker module.
Retrofit slide deck	The Robot Retrofit Kit for Stacker is a slide deck containing detailed retrofit instructions, including CAD images that illustrate the exact placement of each component. This resource also provides a few

Resource	Description
	slides on future Flex robot designs, which incorporate these new safety features into the robot and eliminate the need for retrofitting.